

Homework 3: Market Imperfections

Sergey V. Popov

2026-08-10

Due: Sep 11, 2026, noon.

Asymmetric Cournot

Let there be N firms with costs $TC_i(q_i) = c_i q_i + F$. Let them face the market where the price is $P(Q) = A - BQ$ and Q is the total output of Cournot competitors. Let $A < \max_i c_i$.

- Solve for the Cournot equilibrium price, total quantity, and individual quantity. Notice that generically all q_i s would be different.
- Would all the firms be able to earn positive profits? Which firms are more likely to exit the market?
- Imagine there is a possibility to invest in R&D: lower your c_i but increase your F . Which firms are more likely to do so?

Salop's Model

Imagine a circular model of a city (see Figure MWG 12.C.8 and discussion under it). Firms are symmetric, with zero variable costs and fixed costs F ; the circumference of the city is 1, consumers are distributed uniformly. Consumer that needs to travel distance x to the seller gains utility $v - p - tx$. Let the benefit of purchase be big enough, so that everyone buys something.

- Find the profit of one firm if J firms enter the market. Notice that for each of these firms, the distance to their neighbor from both sides is $1/J$. What happens to the equilibrium price as J changes?
- What J makes profits zero?
- Some firms, say restaurants, demand licensing, increasing the cost F . Why?

Public Good Financing

Let's say each consumer, $i \in \{1..N\}$, has a different preference for the public good:

$$u_i(x_1 + x_2 + \dots + x_N) = \frac{1}{i} \ln(x_1 + x_2 + \dots + x_N) + w_i - x_i$$

- What is the best response of player i to other players' choices of x_{-i} ?
- Argue that only player 1 has $x_1 > 0$, and other players choose $x_i = 0$.
- Argue that maximizing the sum of utilities yields the Pareto optimal allocation.
- How would you achieve the Pareto optimal allocation with market methods?